EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

STRUCTURAL ISOMERISM

1. $CH_3CHOHCH_2CHO$ and $CH_3CH_2CH_2COOH$ constitute a pair of :-
(1) Position isomers (2) Metamers
(3) Optical isomers (4) Functional isomers

SR0001

2. The minimum number of carbon atoms present in an organic compound to show chain isomerism is
(1) 2 (3) 3 (3) 5 (4) 4

SR0002

3. $CH_3—NH—C_2H_5$ and $(CH_3)_3N$ show which type of isomerism :-
(1) Position (2) Functional
(3) Chain (4) None

SR0005

4. $CH_3—\underset{\underset{Cl}{|}}{CH}—CH_2—\underset{\underset{H}{|}}{C}=O$ and

$CH_3—CH_2—CH_2—\underset{\underset{O}{\|}}{C}—Cl$

are constitute a pair of :-
(1) Position isomers
(2) Metamers
(3) Optical isomers
(4) Functional group isomers

SR0006

5. Which are metamers :-
(1) $CH_3–O–CH_2CH_2CH_3$, $CH_3–CH_2–O–CH_2–CH_3$
(2) $C_2H_5–O–C_2H_5$, $CH_3CH_2CH_2CH_2OH$
(3) $CH_3–O–C_2H_5$, $CH_3–CH_2–O–CH_3$
(4) $CH_3–\underset{\underset{O}{\|}}{C}–CH_3$, $CH_3–CH_2–\underset{\underset{O}{\|}}{C}–H$

SR0008

6. Which similarity is necessary for isomerism–
(1) Molecular formula
(2) Structure formula
(3) Physical formula
(4) Chemical formula

SR0009

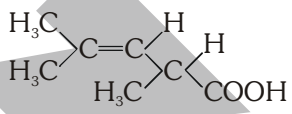
7. $HC\equiv C–CH_2–\underset{\underset{CH_3}{|}}{CH}–CH_3$ & $CH_3–C\equiv C–\underset{\underset{CH_3}{|}}{CH}–CH_3$ are
(1) Chain isomer (2) Homologous
(3) Position isomer (4) None

SR0093

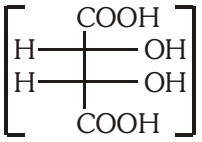
8. How many structural isomer are possible for C_5H_8 having one triple bond ?
(1) 4 (2) 3
(3) 5 (4) 1

SR0094

GEOMETRICAL AND OPTICAL ISOMERISM

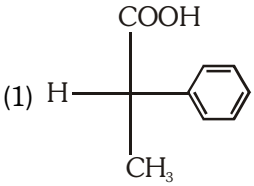
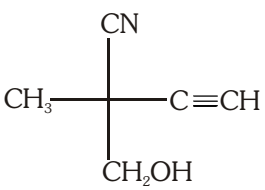
9.  Exhibits :-
(1) Tautomerism
(2) Optical isomerism
(3) Geometrical isomerism
(4) Geometrical and optical isomerism

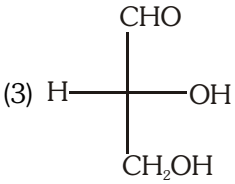
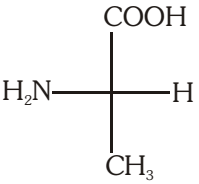
SE0011

10. Meso–tartaric acid  is optically inactive due to the presence of :-
(1) Molecular symmetry
(2) Molecular asymmetry
(3) External compensation
(4) Two asymmetric carbon atoms

SE0019

11. Identify R configuration :

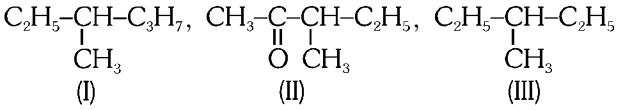
(1)  (2) 

(3)  (4) 

SE0023

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12. Among the following structure I to III



It is true that :-

(1) All three are chiral compounds

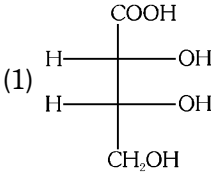
(2) Only I and II are chiral compounds

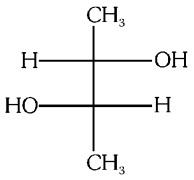
(3) Only II is chiral compound

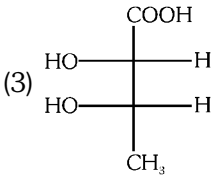
(4) Only I and III are chiral compounds

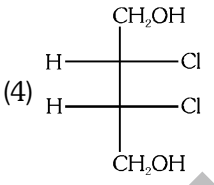
SE0026

13. Which one of the following is a meso–compound.

(1) 

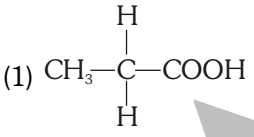
(2) 

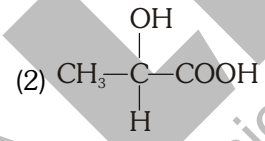
(3) 

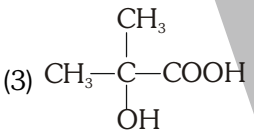
(4) 

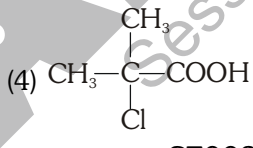
SE0028

14. Which compound is optical active –

(1) 

(2) 

(3) 

(4) 

SE0032

CONFORMATIONAL ISOMERISM

15. Which conformation of butane will have the minimum energy :-

(1) Gauche

(2) Anti/staggered

(3) Eclipsed

(4) None

SE0035

16. Which of the following are true statements.

(a) Alkanes have infinite no. of conformation

(b) The rotation is hindered due to repulsive interaction called torsional strain

(c) The barrier is about 50 kJ/mole

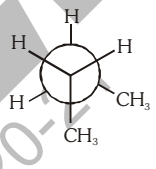
(d) The barrier is about 1-20 kJ/mole

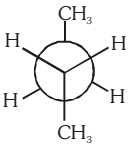
(1) a, b, d (2) a, b, c

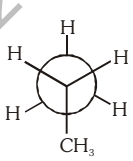
(3) only b (4) only a, d

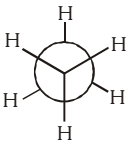
SE0095

17. Which of the following has minimum steric strain?

(1) 

(2) 

(3) 

(4) 

SE0096

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EXERCISE-I (Conceptual Questions) ANSWER KEY

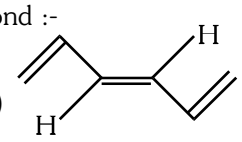
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	2	4	1	1	3	2	2	1	3	2	4	2	2
Que.	16	17													
Ans.	1	4													

EXERCISE-III (Analytical Questions) Master Your Understanding

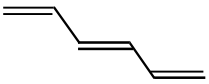
1. Which of the following does not contain any asymmetric carbon but can show enantiomerism:-
(1) Lactic acid (2) 1,3-pentadiene
(3) Tartaric acid (4) 2,3-pentadiene

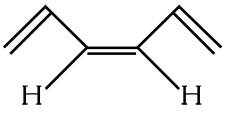
SE0063

2. Which of the following represents the structure having cis arrangement around each double bond :-

(1) 

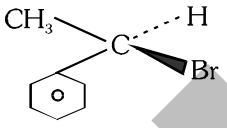
(2) 

(3) 

(4) 

SE0064

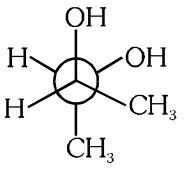
3. The complete IUPAC name of the compound :-

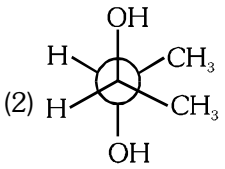


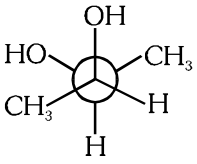
(1) (R)-1-Bromo-1-phenyl ethane
(2) (S)-1-Bromo-1-phenyl ethane
(3) (E)-1-Bromo-1-phenyl ethane
(4) (Z)-1-Bromo-1-phenyl ethane

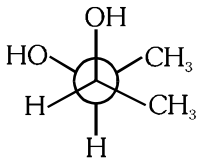
SE0066

4. Which one of the following is the most stable conformation of 2, 3–butanediol :–

(1) 

(2) 

(3) 

(4) 

SE0067

5. How many isomers of $C_5H_{11}OH$ will be primary alcohols (exclude stereoisomers) :–
(1) 2 (2) 3 (3) 4 (4) 6

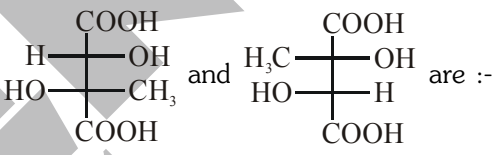
NC0069

6. The minimum number of carbon atoms in ketone to show metamerism :–
(1) 3 (2) 4 (3) 5 (4) 6

SR0070

7. The total number of configurational isomers of the given compound are :-
$CH_3–CH=CHCHOHCHOHCH_3$
(1) 2 (2) 4 (3) 6 (4) 8

SE0073

8.  are :-

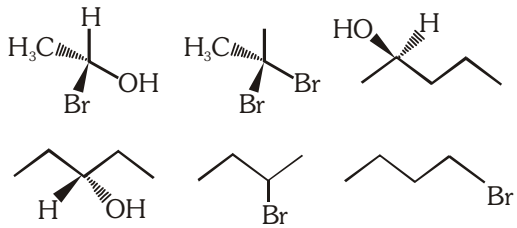
(1) Enantiomers (2) Position isomers
(3) Geometrical isomers (4) Homomers

SE0075

9. Which of the following is not a metamer of $C_4H_{10}O$
(1) Diethyl ether
(2) Methyl n-propyl ether
(3) 2–Methoxy propane
(4) Isobutyl alcohol

SE0076

10. How many compounds among the following are chiral ?



(1) 1 (2) 2
(3) 3 (4) 4

SE0097

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	1	3	3	3	4	4	4	3

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